FY2023 financial statements of 13,956 SMEs

Clustering Question

Goal

- Identify financial profiles of Italian SMEs.
- Explain how these profiles differ economically.
- Examine where fragile profiles are concentrated.

Dataset

Firms	13,956
Distressed firms	1,505
Distress rate	10.78%
Provinces	108
Sectors	306

Key choice: clustering uses financial variables only. Geography, sector, ATECO, and the distress target are used after clustering to interpret the groups.

Financial Variables Used

Financial signal

- Size: revenue and employees.
- Profitability: net and operating margins.
- Cash generation: operating cash flow and OCF margin.
- Debt pressure: financial expenses and Alert Index.
- Productivity: revenue and income per employee.

Preprocessing

- Signed log transforms for skewed monetary variables.
- Winsorization at 0.5th and 99.5th percentiles.
- Median imputation.
- Robust scaling.
- K-means tested for $k = 2, \ldots, 8$.

Chosen solution

$\mathbf{k} = \mathbf{5}$: interpretable financial health ladder

Five SME Financial Profiles

Cluster	Firms	Distress	Net margin	OCF margin
A Resilient high-profit cash generators	2,085	0.43%	15.6%	19.7%
B Healthy profitable core SMEs	6,043	2.75%	4.5%	7.4%
C Thin-margin vulnerable operators	4,745	15.68%	0.7%	3.2%
D Debt-burdened break-even firms	174	31.61%	-1.8%	1.5%
E Loss-making cash-flow distressed firms	909	58.42%	-10.2%	-6.1%

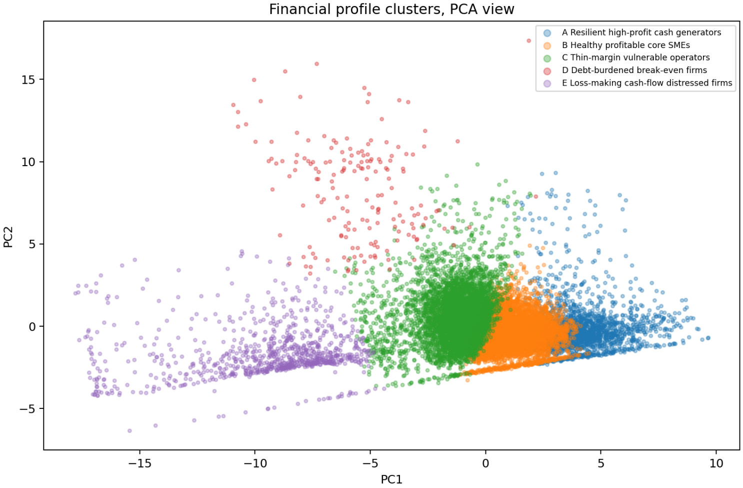
Low-risk side

- Clusters A and B generate profits and cash.
- Distress is below 3% in both profiles.

High-risk side

- Cluster C is the large vulnerable middle.
- Cluster E combines losses and negative cash flow.

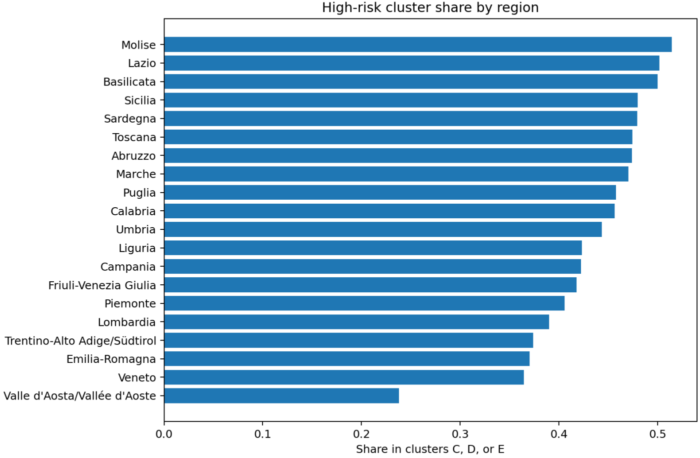
PCA View of the Financial Profiles



Reading the map

- PCA is a linear view of the same scaled financial space used for clustering.
- The profiles form a gradient from profitable cash generators to loss-making firms.
- Distress risk increases as firms move toward weak margins and weak cash flow.

Geography of Fragile Profiles



High-risk profiles

- C: thin-margin vulnerable
- D: debt-burdened break-even
- E: loss-making cash-flow distressed

Region	High-risk share
Molise	51.43%
Lazio	50.17%
Basilicata	50.00%
Sicilia	48.01%
Sardegna	47.96%

This is a concentration pattern, not a causal claim about location.

Main Takeaways

1. The clusters are economically meaningful

The k=5 solution separates SMEs by profitability, cash generation, productivity, and debt-service pressure.

2. Distress follows the financial health ladder

Distress rises from 0.43% in the strongest cluster to 58.42% in the loss-making cash-flow cluster.

3. Geography and sector enrich the interpretation

Fragile profiles are not uniformly distributed: some regions and activities contain a larger share of thin-margin, debt-burdened, or loss-making firms.

The clustering turns financial statements into an interpretable risk map.